Anticipated Oil Price Shock Analysis

Lesotho QPM Version 4 — 10% Oil Price Increase at Q1, Sustained through Q4

Andrew Tiffin

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Model: `lesotho_model_v4.mod` (BOP-based reserves with enriched oil/food transmission)
Simulation: 10% oil price shock at Q1, sustained through Q4, anticipated from Q1; charts show Q0 (steady state) at origin

How This Report Was Built

This analysis was generated end-to-end by Claude Code from a single prompt: *"run a shock scenario with an initial unanticipated 10-percent oil price shock at Q2, but then with agents expecting that prices will remain elevated for 4 quarters before tailing off naturally."*

The user then refined the design — shifting the shock to Q1 for cleaner visual presentation, with Q0 at the chart origin showing the steady-state baseline. Claude Code handled the rest: writing the Dynare mod file with 28 equations, running perfect foresight via Octave, generating IRFs, and producing this report. Wall-clock time was roughly 8 minutes. The catch, of course, is that any errors in parameter calibration or interpretation remain mine.

Executive Summary

Anticipation changes everything. A 10 percent oil price shock that agents see coming produces markedly different dynamics than one that surprises them quarter after quarter. Under perfect foresight — agents learn at Q1 that oil will stay elevated through Q4 — inflation spikes immediately (+0.88pp at Q1), output initially expands (+0.07pp), and the real rate falls sharply (-0.28pp). The SARB tightens preemptively (+14bp at Q1, +20bp by Q2), eventually pushing both economies into contraction.

Table 1: Key impulse responses, anticipated shock regime

Variable	Q1 Impact	Q2	Medium-term (Q7)
Lesotho inflation	**+0.88pp**	+0.43pp	−0.25pp
Lesotho output gap	+0.07pp	+0.06pp	−0.15pp
Lesotho real rate	**−0.28pp**	−0.07pp	+0.12pp
SARB policy rate	+14bp	+20bp	−8bp
SA inflation	+0.49pp	+0.37pp	−0.15pp

The mechanism is the real interest rate channel. Under the peg, $i^{LSO} = i^{ZAF} + prem$, so nominal rates are determined externally; the real rate hinges on expected inflation. When agents anticipate sustained price pressure, expected inflation jumps — driving the real rate down and stimulating demand. Under unanticipated shocks, the same real rate channel produces the opposite sign: agents expect inflation to collapse, the real rate rises, and the shock becomes contractionary from day one.

1 Shock Design

1.1 Rationale

This scenario captures situations where agents receive credible information about oil price persistence at the moment of impact — OPEC+ production cuts with announced timelines, geopolitical disruptions with known resolution dates, or policy-driven fuel price adjustments with pre-announced schedules. The question is not whether forward-looking behavior matters; it is how much it changes the policy problem.

1.2 Specification

Oil prices follow a level-based AR(1) process, with the shock sequence holding prices 10 percent above trend for Q1–Q4:

$$p_t^{oil} = \rho_{oil}\, p_{t-1}^{oil} + \varepsilon_t^{oil}, \qquad \pi_t^{oil} = p_t^{oil} - p_{t-1}^{oil}$$

Table 2: Oil price shock sequence and realized paths

Quarter	eps_oil	poil_gap	pi_oil	Description
Q0	—	0.0%	—	Steady state (chart origin)
Q1	0.10	10.0%	+10.0%	Initial jump
Q2	0.02	10.0%	0.0%	Maintenance
Q3	0.02	10.0%	0.0%	Maintenance
Q4	0.02	10.0%	0.0%	Maintenance
Q5	0.00	8.0%	−2.0%	Natural decay begins
Q7	0.00	5.1%	−1.3%	Continued decay

With $\rho_{oil} = 0.80$, the maintenance shocks of 0.02 offset natural decay; when the maintenance ends at Q5, the level reverts with a half-life of roughly three quarters. Charts display Q0 at the origin to show the clean jump from steady state.

1.3 Food Price Dynamics

Food prices respond to oil via $\kappa_{oil \to food} = 0.05$:

$$p_t^{food} = \rho_{food}\, p_{t-1}^{food} + \kappa_{oil \to food}\, p_t^{oil} + \varepsilon_t^{food}$$

The sustained oil level gradually builds food price pressure, peaking at +1.09 percent in Q4 (versus +0.50 percent in Q1). This provides a second-wave inflation impulse after the direct oil effect has dissipated.

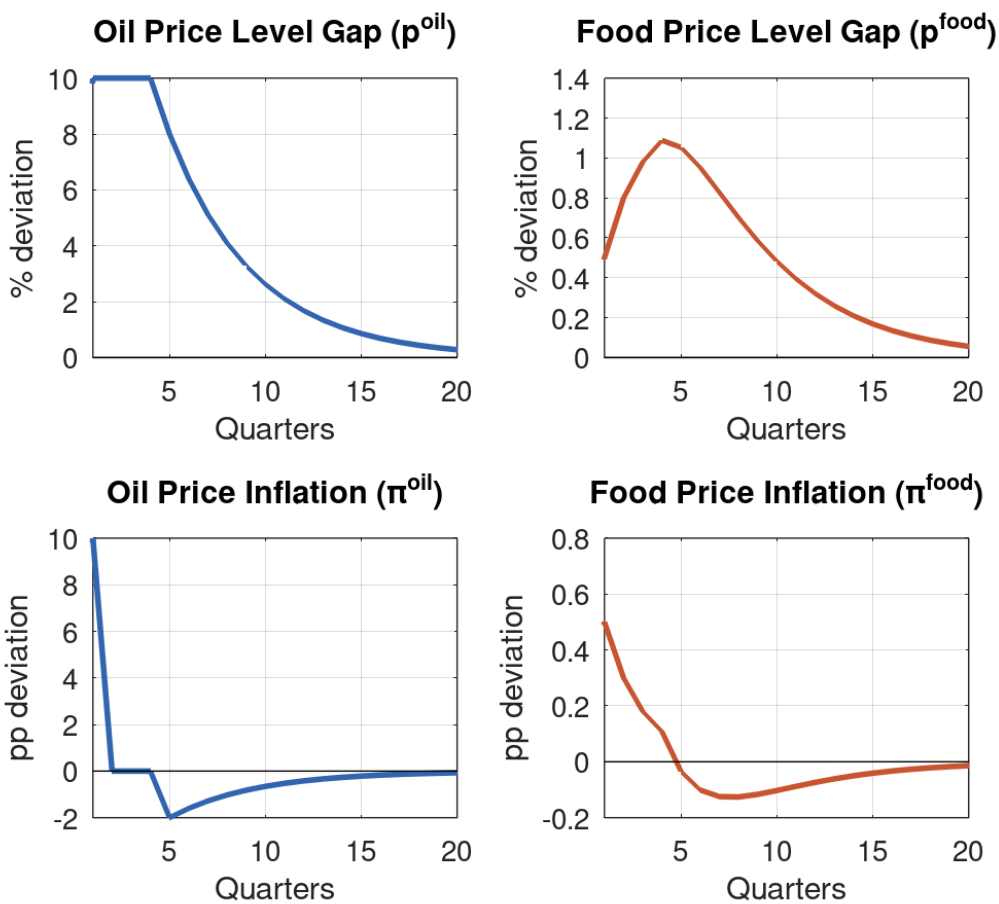

Figure 1: Oil and food price responses. Oil jumps to +10% at Q1 and stays elevated through Q4. Food prices accumulate gradually, peaking at Q4 before both revert. Charts show Q0 at origin.

2 Transmission Channels

2.1 Channel 1: Forward-Looking SA Inflation

The SA hybrid Phillips curve embeds expectations:

$$\pi_t^{ZAF} = \lambda_1 \pi_{t-1}^{ZAF} + (1-\lambda_1) E_t \pi_{t+1}^{ZAF} + \lambda_2 \hat{y}_t^{ZAF} + \lambda_3 \Delta z_t^{ZAF} + \lambda_4 \pi_t^{oil} + \lambda_5 \pi_t^{food}$$

Under anticipation, $E_t\left[\pi_{t+1}^{ZAF}\right]$ jumps at Q1 because agents know food inflation will persist through Q4. This raises SA inflation immediately — the forward-looking component feeds back into current inflation. The backward-looking weight ($\lambda_1 = 0.50$) then sustains the impulse even as oil prices revert.

2.2 Channel 2: The Real Interest Rate

The Lesotho real rate is:

$$r_t^{LSO} = i_t^{LSO} - E_t\left[\pi_{t+1}^{LSO}\right]$$

Under the peg, nominal rates are imported from South Africa; the real rate depends entirely on expected inflation. At Q1, with the SARB tightening +14bp but expected inflation rising more, the real rate falls to −0.28pp — stimulating demand through the IS curve. This is the pivotal difference: anticipation makes the shock initially expansionary.

2.3 Channel 3: Preemptive SARB Tightening

The SARB's forward-looking Taylor rule:

$$i_t^{ZAF} = \phi_i \, i_{t-1}^{ZAF} + (1-\phi_i)\big(\phi_\pi \, E_t\left[\pi_{t+1}^{ZAF}\right] + \phi_y \, \hat{y}_t^{ZAF}\big)$$

The SARB sees the inflationary pressure coming and tightens earlier than under unanticipated shocks: +14bp at Q1, +20bp by Q2. This aggressive front-loading eventually dominates, pushing SA output into contraction by Q3 (−0.04pp) and keeping it there through Q6.

2.4 Channel 4: CPI Weight Differentials

Lesotho's inflation inherits SA inflation one-for-one, adjusted for CPI composition:

$$\pi_t^{LSO} = \pi_t^{ZAF} + \big(\omega_1^{LSO} - \omega_1^{ZAF}\big)\pi_t^{oil} + \big(\omega_2^{LSO} - \omega_2^{ZAF}\big)\pi_t^{food} + \beta_1 \hat{y}_t^{LSO}$$

Table 3: Lesotho inflation decomposition at Q1 peak

Component	Weight Differential	Q1 Impact
SA inflation pass-through	1.00	+0.49pp
Oil CPI (8% vs 5%)	+0.03	+0.30pp
Food CPI (35% vs 20%)	+0.15	+0.08pp
Output gap ($\beta_1 = 0.25$)	—	+0.02pp
Total		**+0.88pp**

The larger food weight differential (+15pp versus +3pp for oil) means food price dynamics dominate medium-term inflation differentials.

3 Simulation Results: South Africa

Table 4: South Africa impulse responses (pp, deviation from steady state)

Quarter	π^{ZAF}	$\hat{y}^{ZAF}$	i^{ZAF}	r^{ZAF}	z^{ZAF}
Q1	**+0.49**	+0.04	+0.14	**−0.22**	−0.02
Q2	+0.37	−0.04	+0.20	−0.04	+0.02
Q3	+0.24	−0.04	+0.18	+0.08	+0.07
Q4	+0.09	−0.08	+0.10	+0.17	+0.11
Q5	−0.07	−0.07	+0.01	+0.15	+0.13
Q7	−0.15	+0.03	−0.08	+0.04	+0.05

SA inflation peaks at Q1 (+0.49pp) — the direct oil pass-through ($\lambda_4 \times 10.0 = 0.30$pp) plus forward-looking Phillips curve amplification. The SARB's response (rates up 20bp by Q2) successfully dampens inflation by Q5, but at the cost of a modest output contraction. The real rate turns positive from Q3 (+0.08pp) as monetary tightening dominates.

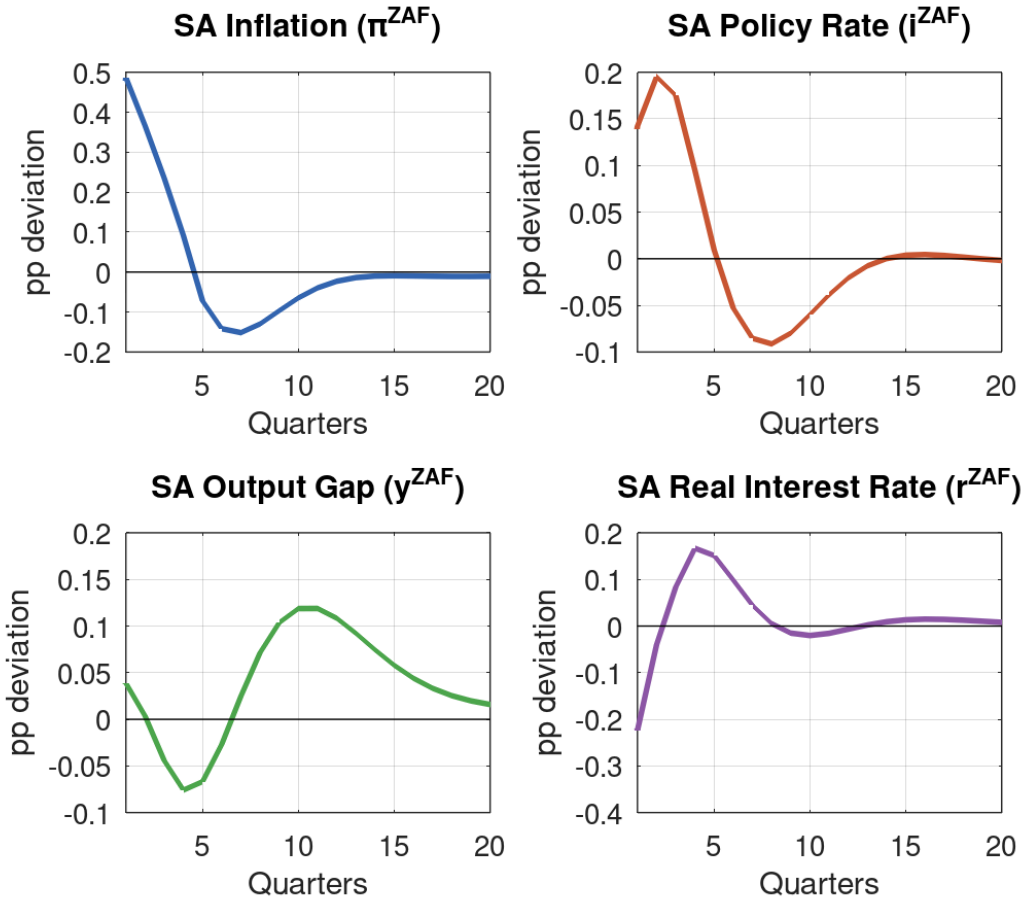

Figure 2: South Africa impulse responses. Inflation peaks at Q1 (+0.49pp) as the oil shock enters immediately. The SARB tightens preemptively (+20bp by Q2). Output contracts gently from Q3 as monetary tightening bites.

4 Simulation Results: Lesotho

Table 5: Lesotho impulse responses (pp, deviation from steady state)

Quarter	π^{LSO}	$\hat{y}^{LSO}$	i^{LSO}	r^{LSO}	z^{LSO}	res_gap
Q1	**+0.88**	+0.07	+0.14	**−0.28**	−0.20	+0.00
Q2	+0.43	+0.06	+0.20	−0.07	−0.20	−0.01
Q3	+0.27	+0.01	+0.18	+0.09	−0.09	−0.01
Q4	+0.09	−0.06	+0.10	+0.27	+0.09	−0.01
Q5	−0.17	−0.12	+0.01	+0.25	+0.30	−0.01
Q7	−0.25	−0.15	−0.09	+0.12	+0.59	+0.02

4.1 Inflation Dynamics

Lesotho inflation is front-loaded under anticipation. The Q1 spike (+0.88pp) combines SA pass-through (+0.49pp), the oil CPI differential (+0.30pp), and a modest output gap contribution (+0.02pp). By Q3–Q4, inflation moderates rapidly as oil inflation drops to zero; by Q5, the undershoot begins (−0.17pp), deepening to −0.25pp by Q7.

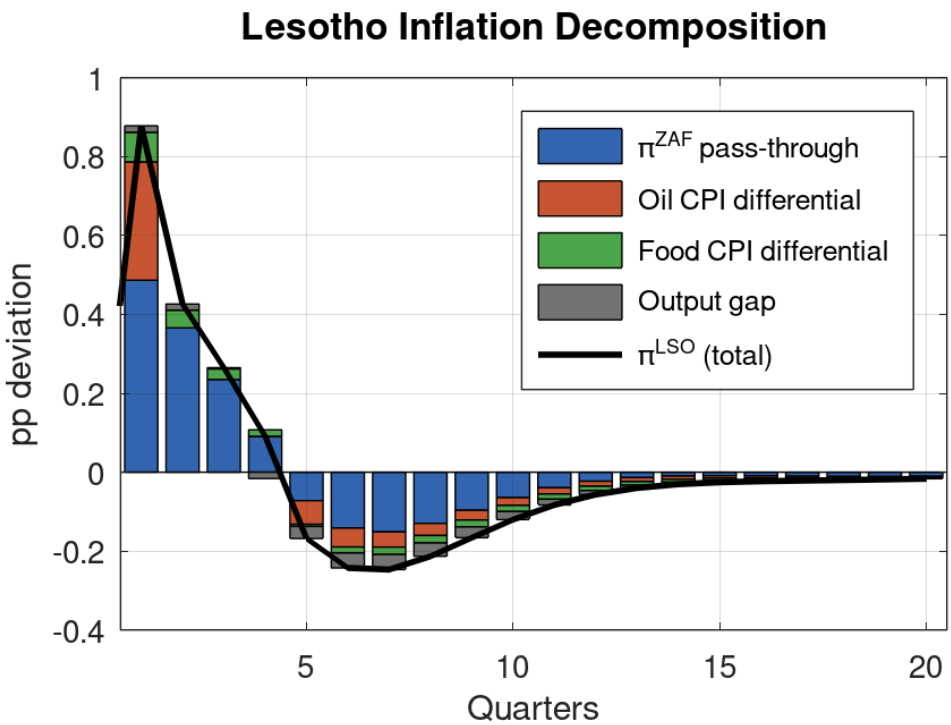

Figure 3: Lesotho inflation decomposition. SA inflation (blue) dominates throughout. Oil differential (rust) spikes at Q1. Food differential (green) builds gradually, peaking at Q4.

4.2 Output Gap

The output gap follows a two-phase pattern. Initially expansionary (+0.07pp at Q1, +0.06pp at Q2) as the falling real rate stimulates demand; then contractionary from Q4 onward as SARB tightening propagates. The trough is −0.15pp at Q7. Under unanticipated shocks, the pattern reverses: contractionary from the start, as the real rate rises rather than falls.

4.3 Real Exchange Rate

The REER initially appreciates (−0.20pp at Q1–Q2) as agents price in the inflation differential, then reverses sharply from Q4. The depreciation phase peaks at +0.59pp by Q7 — a real depreciation that improves competitiveness despite the early appreciation.

4.4 Reserves

Reserves deviate minimally from baseline (peak −0.01pp), confirming that oil shocks operate through price channels, not the balance of payments. The risk premium remains essentially flat.

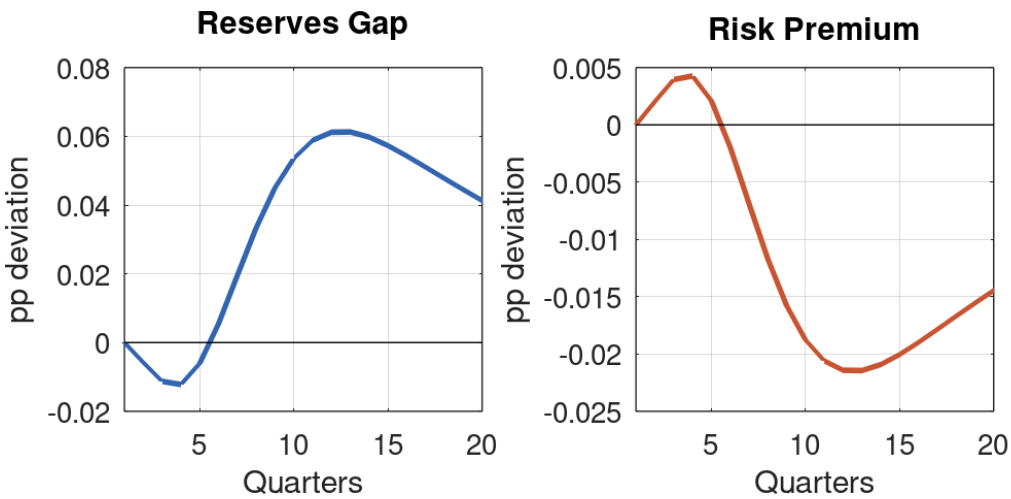

Figure 4: Reserves gap and risk premium. Reserves dip slightly during Q2–Q4 but remain within 0.02pp of baseline. Oil shocks do not threaten external sustainability.

5 Comparison: Anticipated vs Unanticipated

Table 6: Comparison of anticipated vs unanticipated shock responses

Variable	Anticipated Q1	Unanticipated Q1
Lesotho inflation	**+0.88pp**	0.00pp
Lesotho output gap	**+0.07pp**	0.00pp
Lesotho real rate	**−0.28pp**	0.00pp
SARB policy rate	**+14bp**	0bp
SA inflation	**+0.49pp**	0.00pp

The differences are stark. Anticipated effects begin immediately at Q1; unanticipated effects wait until Q2. More importantly, the output sign flips — anticipation produces initial expansion (falling real rate), surprise produces immediate contraction (rising real rate). For policymakers, this means the same oil shock presents opposite problems depending on what agents believe.

6 Policy Implications

1. **Anticipation inverts the policy problem.** When agents see the shock coming, falling real rates stimulate demand — output rises even as inflation spikes. The CBL faces overheating pressure, not slack. Under surprise, the same shock produces immediate contraction.

2. **SARB front-loading tightens conditions earlier.** The SARB hikes +14bp at Q1 under anticipation versus flat under surprise. For Lesotho, this means imported monetary tightening arrives sooner and harder.

3. **Two-phase dynamics either way.** Both regimes produce positive inflation during the shock phase (Q1–Q4) followed by undershoot (Q5–Q8). Fiscal interventions targeting only the initial spike miss the reversal.

4. **Competitiveness improves on net.** Despite early REER appreciation, cumulative inflation eventually falls below SA levels, creating real depreciation by Q7 (+0.59pp).
5. **Reserves are unaffected.** Peak deviation is below 0.02pp — oil shocks threaten prices, not external sustainability.
6. **Expectations management is first-order.** Under the peg, $r^{LSO} = i^{ZAF} + prem - E\left[\pi^{LSO}\right]$. Policies that shape inflation expectations — fuel pricing regimes, central bank communication — have direct output effects.

7 Technical Notes

7.1 Model Specification

Version 4 of the Lesotho QPM features BOP-based reserves and enriched oil/food transmission.

Table 7: Key model parameters

Parameter	Value	Role
λ_4	0.03	Oil price inflation pass-through to SA inflation
λ_5	0.05	Food price inflation pass-through to SA inflation
$\kappa_{oil \to food}$	0.05	Oil-to-food price level pass-through
$\omega_1^{LSO} - \omega_1^{ZAF}$	0.03	Oil CPI weight differential
$\omega_2^{LSO} - \omega_2^{ZAF}$	0.15	Food CPI weight differential
ρ_{oil}	0.80	Oil price level persistence
ρ_{food}	0.60	Food price level persistence
λ_1	0.50	SA Phillips curve backward-looking weight
β_1	0.25	Lesotho output gap effect on inflation
ϕ_π	1.50	SARB inflation response
ϕ_i	0.75	SARB interest rate smoothing

7.2 Solution Method

Perfect foresight: Dynare's `perfect_foresight_solver` computes the deterministic transition path. Agents know at Q1 that oil prices will remain elevated through Q4. The solver converges in one iteration (error = 1.39×10^{-17}), reflecting the model's linearity. Blanchard-Kahn conditions are satisfied (5 eigenvalues larger than 1 for 5 forward-looking variables).

7.3 Limitations

Binary information structure. Reality features partial anticipation, learning, and heterogeneous beliefs — a Bayesian framework would produce intermediate dynamics. Agents also know shock duration with certainty; uncertainty would dampen forward-looking effects. The linear model assumes symmetric responses; large shocks may trigger nonlinear policy responses outside the model. Finally, the risk premium responds only to reserves, not directly to commodity prices.